

Lab Notebook (Microfluidic Platform)

June

Built and validated the hardware signal chain for microfluidic fluorescence detection and droplet sorting: the PMT detects fluorescence and outputs weak pulses, which are converted into digital pulses by the TLV3501 comparator; the PYNQ-Z2 FPGA identifies target droplets and outputs TTL trigger signals, driving the function generator to produce low-voltage sine bursts, which are then amplified by the high-voltage amplifier to drive the DEP electrodes for droplet sorting. Feasibility was verified separately for the PMT-comparator, FPGA-function generator, and overall timing alignment.

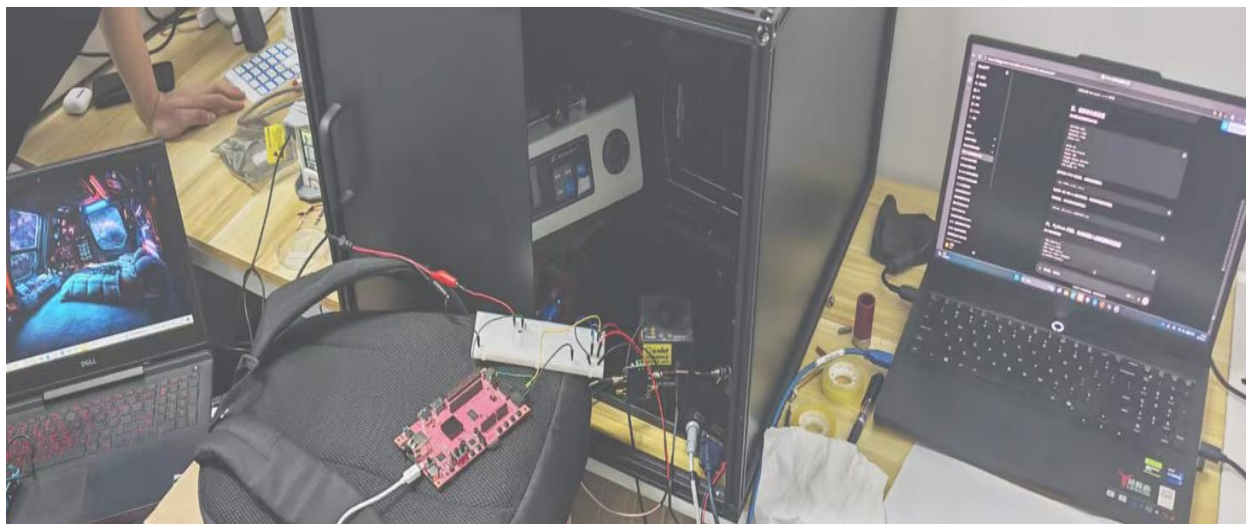
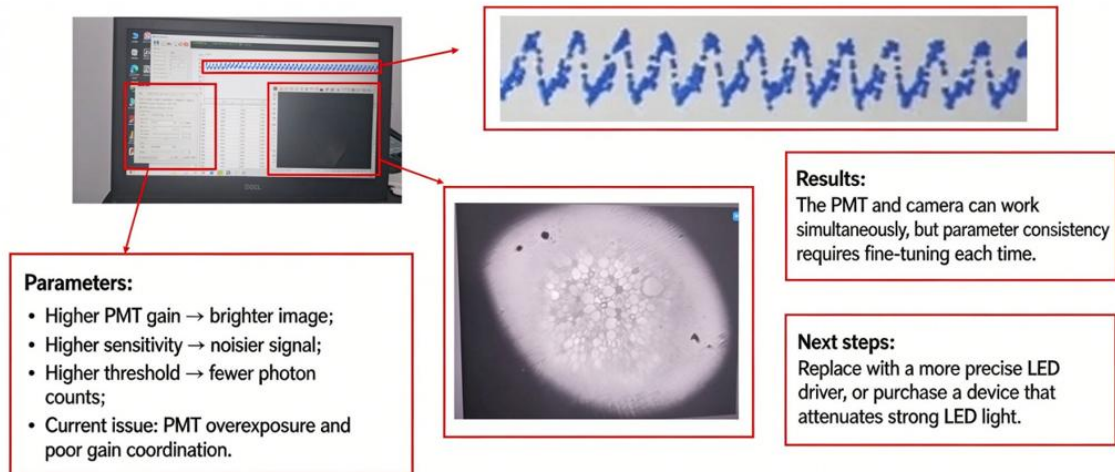
Optical-Path Modification Test

Completed PMT-comparator and FPGA-function generator verification (LED + PMOD rising-edge triggering, observed on an oscilloscope).



Video 5 — PMT-comparator and FPGA-function generator verification

Built the complete hardware platform for droplet microfluidic sorting, completing droplet generation, optical detection, and circuit-chain debugging. The PMT and camera can work simultaneously, but the parameters are difficult to unify and require fine-tuning each time.



The FPGA signal response had faults, and the optical and circuit parameters were difficult to match; subsequent plans include optimizing the optical path, rewriting the FPGA program, and conducting high-voltage amplifier validation tests.

First Attempt at Droplet Sorting

Droplet generation is uniform, stable, and efficient.



Video 6 — Uniform, stable droplet generation

After optimizing the droplet-sorting control signal chain, the continuous sorting function was tested for the first time.

The sorting channel clogs easily, and oversized droplets cannot enter the sorting channel.



Video 7 — Droplet sorting attempt

All videos referenced in this document are available for download at the bottom of our team's Notebook page.